

- 4 When seeds germinate, starch is broken down and used as a respiratory substrate to provide energy for growth.

Respiration rate can be measured using a simple respirometer.

A student knew that mung beans germinate very quickly and wanted to compare the respiration rate in germinating mung beans with the rate in germinating peas.

They formed the following hypothesis:

Germinating mung bean seeds will have a faster respiration rate than germinating pea seeds.

Plan an investigation to find evidence to support or reject this hypothesis.

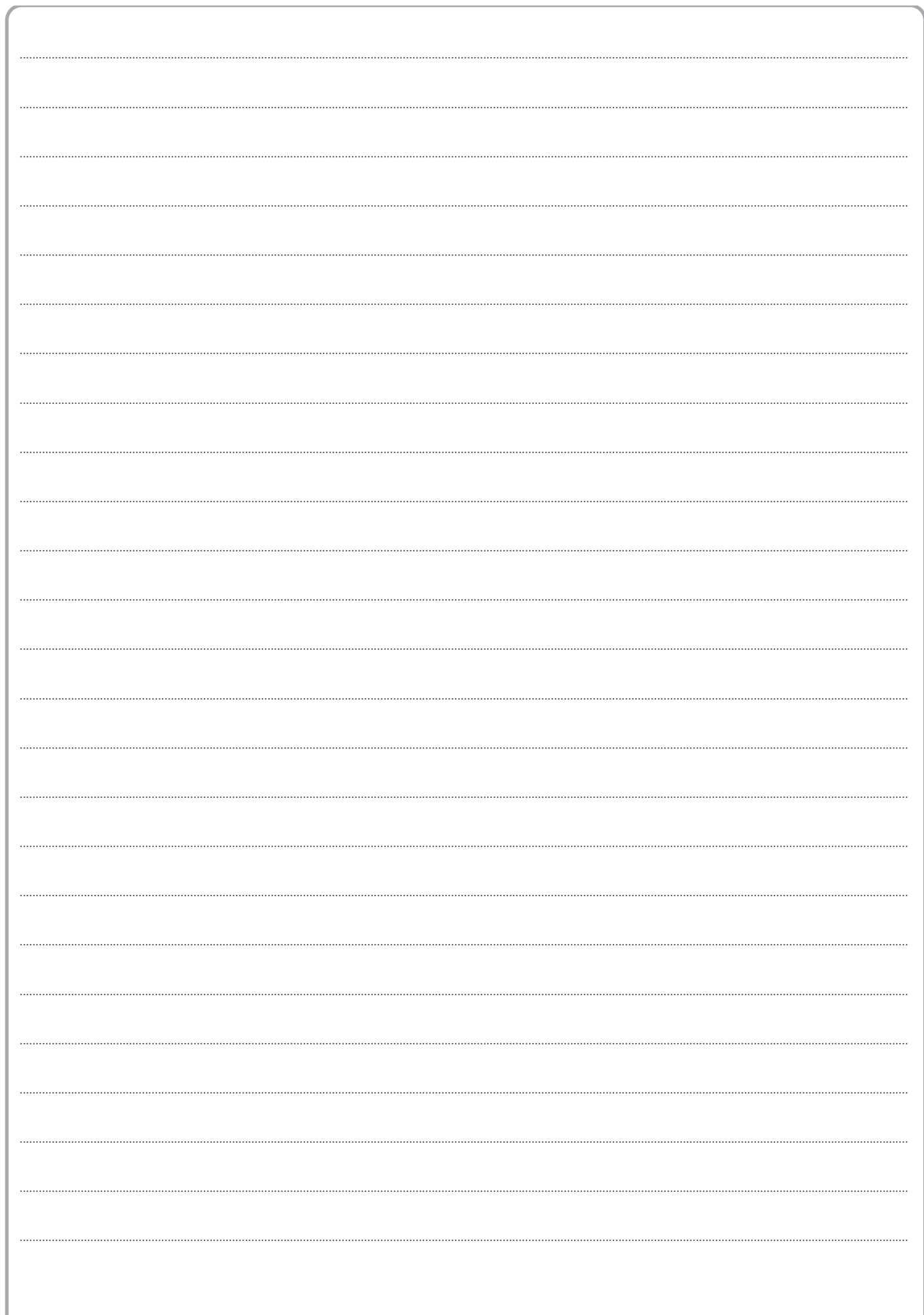
(a) Describe preliminary practical work that you might undertake to ensure your proposed method would provide quantitative results.

(3)

(b) Devise a detailed method, including how you would control and monitor important variables.

(8)

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(c) Describe how your results should be recorded, presented and analysed in order to draw conclusions from your investigation.

(3)

(d) Suggest **three** limitations of your proposed method.

(3)

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(Total for Question 4 = 17 marks)

TOTAL FOR PAPER = 50 MARKS